

EE 623P: Practicum on Digital Control of Power Electronics and Drives

Course Code	EE 623P
Course Name	Practicum on Digital Control of Power Electronics and Drives
L-T-P-C	1-0-4-3
Prerequisites	EE504, EE508, EE527
Intended for	UG and PG
Distribution	Core for M. Tech (PED), Elective for UG and other PG courses
Approval	19th BoA (Nov 2017); 15th Senate (Feb 2018)
Revised	23rd Senate (Oct 2019) - renamed from "Practicum on Digital Control of Electrical Drives"; 36th BoA (Sep 2020) - revised

Course Contents

- **Introduction to digital control, microprocessor/FPGA**

Lecture: Why digital control, challenges in digital platform, fixed-point and floating point representations, sampling, mathematical modelling of sampling, quantization and its impact on stability, s-domain to z-domain mapping (tustin, forward difference, backward difference), stability analysis in z-domain, discretization of analog controllers. [5 Lectures]

Laboratory: Implementation of various building blocks (ADC sampling, PWM block, controller block) required for closed-loop control of power electronic converters (using C/Verilog coding). [20 Hours]

- **Closed-loop control of DC-DC converters**

Lectures: Controller design techniques, discretization of analog controllers, different types of delays introduced by the digital controllers. [5 Lectures]

Laboratory: Implementation of digital voltage mode control, load transient performance, soft start-up, anti-windup arrangement. [20 Hours]

- **Module III**

Lecture: Digital Proportional-Resonant and Proportional Integral controller design for a single-phase and a three-phase voltage source inverter [2 Lectures]

Laboratory: Implementation of a single-phase voltage source inverter with different loads (standalone case), digital controller design and implementation of different modulation techniques in a three-phase voltage source converter [8 Hours]

- **Module IV**

Lecture: Design of a digital controller in a buck fed DC motor [1 Lectures]

Laboratory: Digital implementation of closed-loop control [4 Hours]

- **Module V**

Lecture: Digital V/F controller design in an induction machine [1 Lectures]

Laboratory: Implementation of digital V/F control [4 Hours]

Text Books

1. B. C. Kuo, **Digital Control Systems**, Oxford University Press, 2012.
2. N. Mohan, T. M. Undeland, and W. P. Robbins, **Power Electronics Converters, Applications, and Design**, 3rd edition, Wiley India, 2008.

References

1. S. N. Vukosavic, **Digital Control of Electrical Drives**, springer, 2017.
2. R. W. Erickson and D. Maksimovic, **Fundamentals of Power Electronics**, 2nd edition, Dordrecht, The Netherlands: Kluwer, 2001.